

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A MCQ TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I __Gowtham kumar /192525163__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Gowtham Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Knowledge Discovery: From Raw Data to Smart Decisions



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INTRODUCTION



- ▶ Knowledge Discovery in Databases (KDD) is the process of extracting useful and meaningful knowledge from large volumes of data.
- ▶ It transforms raw data into valuable insights that support better decision-making.
- ▶ The KDD process includes Data Cleaning, Data Integration, Data Selection, Data Transformation, Data Mining, Pattern Evaluation, and Knowledge Presentation.
- ▶ It helps identify hidden patterns, trends, and relationships that are difficult to detect through manual analysis.
- ▶ KDD is widely used in banking, healthcare, e-commerce, education, and social media for data-driven decision-making.

PROJECT OBJECTIVES



- ▶ To understand how raw data is transformed into meaningful knowledge using the KDD process.
- ▶ To explore the complete journey from data collection to smart decision-making.
- ▶ To learn the seven key stages of Knowledge Discovery in Databases (KDD) and their importance.
- ▶ To identify hidden patterns, trends, and valuable insights from large datasets.
- ▶ To understand the role of Data Mining in discovering useful information from data.

PROBLEM STATEMENT



- ▶ The Challenge of Extracting Valuable Knowledge from Massive Data
- ▶ Every day, organizations generate huge volumes of data from online transactions, social media, healthcare, banking, and e-commerce.
- ▶ Most of this data is raw, unorganized, and difficult to interpret, making manual analysis nearly impossible.
- ▶ Important patterns and relationships often remain hidden within large datasets.
- ▶ Poor-quality data, such as missing, duplicate, or inconsistent records, reduces the accuracy of analysis.

PROPOSED SOLUTION

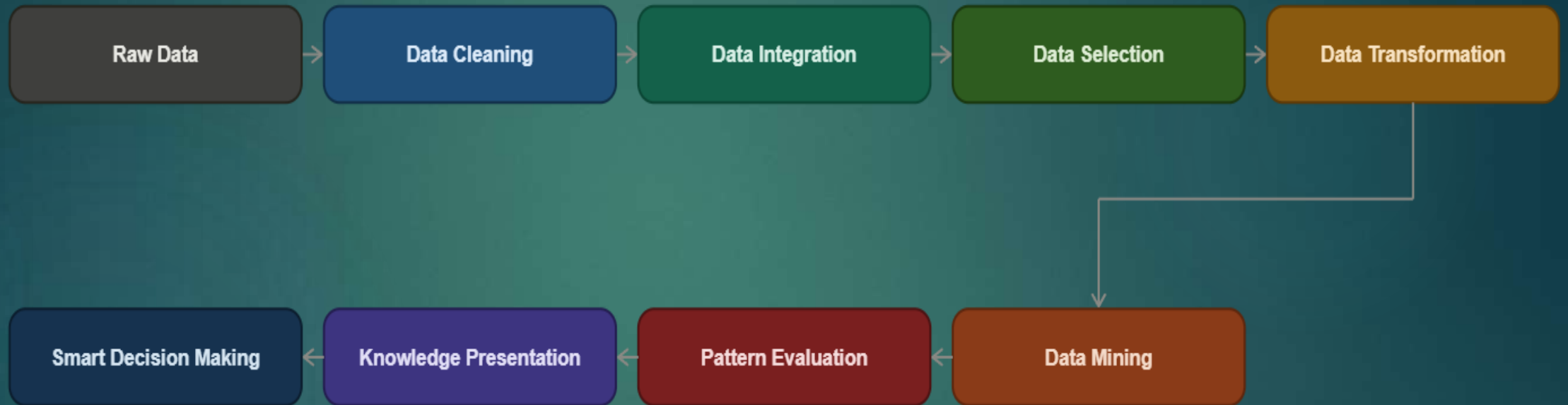
- ▶ **Knowledge Discovery in Databases (KDD)**

- ▶ KDD provides a structured process to convert raw data into valuable knowledge.

Key Stages

- ▶ Data Cleaning
- ▶ Data Integration
- ▶ Data Selection
- ▶ Data Transformation
- ▶ Data Mining
- ▶ Pattern Evaluation
- ▶ Knowledge Presentation
- ▶ **Outcome** : Useful patterns and insights that support informed business decisions.

SYSTEM DESIGN



RESULTS

▶ **Expected Outcomes**

- ▶ The KDD process successfully:
- ▶ Converts raw data into meaningful information.
- ▶ Identifies hidden patterns and relationships.
- ▶ Supports accurate predictions and decision-making.
- ▶ Improves efficiency and business intelligence.

▶ **Example**

- ▶ An online shopping platform uses KDD to recommend products based on customers' purchase history, increasing customer satisfaction and sales.

CONCLUSION

Knowledge Discovery in Databases (KDD) is a systematic process that transforms raw data into meaningful and valuable knowledge.

The KDD process helps uncover hidden patterns, trends, and relationships that support informed decision-making.

Data Mining is the core stage of KDD, responsible for extracting useful patterns from large datasets.

KDD is widely used in banking, healthcare, e-commerce, education, and business analytics to improve efficiency and solve real-world problems. As data continues to grow,

KDD will remain a key technology for driving innovation and intelligent decision-making in the future.